

FalCom: An MCMC Sampling Framework for Facility Location and Districting Problems

Alaittin Kırtışoğlu¹ and Hemanshu Kaul¹

Abstract

We introduce *FalCom*, the first Markov chain Monte Carlo sampler for hierarchical districting with capacitated facility location. FalCom partitions a geographic region into balanced and contiguous districts at multiple administrative levels and assigns capacitated facilities at each level, with the number of districts emerging from capacity and demand. Each proposal cuts spanning trees of merged subgraphs, so contiguity holds structurally at every level.

Distributional convergence is reported empirically via the Gelman-Rubin $\hat{R}$ statistic and effective sample size diagnostics on synthetic grids up to fifty thousand basic units. For small datasets, we define an objective function and use FalCom as an optimizer to compare its result with the solutions of a MILP formulation of the problem. FalCom finds a near-optimum solution with a gap below 1% for these experiments, validating its performance against the MILP optimum. For larger datasets, we deploy FalCom on the London Ambulance Service for ensemble analysis to expose which boundaries are forced by geography and demand, which stations are essential, and where the current operational layout sits among feasible alternatives.

1 Introduction

Districting is the problem of grouping small geographic areas, called *basic units*, into larger geographic clusters, called *districts*, such that the resulting partition satisfies three classical planning criteria: *balance*, *contiguity*, and *compactness* [29]. Balance requires districts of equitable size with respect to one or more activity measures such as workload, sales potential, population, or demand. A district is contiguous if one can travel between any two of its basic units without leaving the district, and it is compact if it is somewhat round-shaped and undistorted. Depending on the practical context, districting is also called territory design, territory alignment, zone design, or sector design [30]. In addition, for each district often the location of a *facility* is either given or should be sought: a branch office, a depot, a clinic, or the home address of a service person [29].

Districting problems arise in a broad range of domains. Political redistricting re-partitions a jurisdiction into electoral constituencies with near-equal populations [45]. Sales territory design assigns customers to sales representatives under workload balance and travel-time constraints [47, 63]. Service districting defines catchment areas for social facilities such as hospitals, schools, police patrols, and home-care teams [9, 11, 16]. Distribution districting creates delivery zones for postal services, waste collection, and meter reading [19, 40, 52]. Despite this diversity, all four classes share a common mathematical core: partitioning a spatial graph into connected, capacity-respecting subgraphs.

The quality of a districting plan has direct consequences for both the populations served and the organizations that serve them. Poorly drawn districts create unbalanced workloads: some service

¹Department of Applied Mathematics, Illinois Institute of Technology, Chicago, IL 60616. E-mail: kaul@iit.edu, kirtisoglu@hawk.iit.edu

teams are overwhelmed while others are underutilized, degrading response times and increasing operational costs. Non-contiguous or non-compact districts force longer travel routes for sales persons, delivery vans, patrol officers, and home-care nurses, raising day-to-day expenses [29]. In healthcare, district boundaries determine which communities a primary-care team serves; a poorly designed partition can leave vulnerable populations far from the nearest facility, undermining equitable access. In emergency services, districts define response zones whose shapes directly affect survival rates. Beyond service delivery, every districting incurs administrative transition costs, such as retraining staff, reassigning caseloads, updating information systems, and coordinating across bureaucratic boundaries, so frequent or drastic redesigns are costly [30]. These stakes motivate the search for principled methods that can explore the space of feasible plans systematically, rather than committing to a single design based on incomplete information.

Although most of the location science literature focuses on where to locate facilities, it has been observed that “equally important is the second aspect of location problems that is usually not mentioned explicitly: the allocation of customers to facilities” [29]. The focus of districting inverts the classical perspective: first determine which customers should be served together, then locate facilities to serve them. This observation reveals a deep connection between districting and capacitated facility location. In political redistricting, “facilities” are elected representatives, each serving one district; in capacitated facility location, facilities serve catchment areas subject to demand constraints. Techniques developed for one problem transfer naturally to the other whenever contiguity and capacity play central roles.

Many real-world systems require districts at multiple administrative levels. In healthcare networks, primary-care districts nest inside referral regions; in public administration, municipalities nest inside counties or provinces. We call these higher-level aggregations *superdistricts*. A hierarchical districting problem simultaneously partitions basic units into districts and districts into superdistricts, subject to balance and contiguity constraints at every level. The hierarchical capacitated facility location problem (HCFL) generalizes this further by requiring that a facility be assigned at each level, with capacity constraints coupling the levels [14].

Districting problems are computationally challenging. Enforcing contiguity requires each district to induce a connected subgraph, which introduces an exponential number of constraints in integer programming formulations [54, 59]. Adding strict capacity bounds further shrinks the feasible region, and multi-level hierarchy compounds the combinatorial explosion. As a result, exact mixed-integer programming methods are generally limited to instances with a few hundred to one thousand basic units [48, 51]. Metaheuristic approaches, including GRASP, tabu search, simulated annealing, and genetic algorithms, extend this range to roughly 1,000–5,000 units [10, 16, 47], but they provide no distributional guarantees on the quality of sampled solutions. A comprehensive survey [29] concludes that “research should first and foremost concentrate on a common and generic framework for districting problems” rather than continue developing application-specific heuristics with yet another compactness measure or balance constraint. This paper answers that call.

A promising direction emerged from computational political redistricting. The ReCom (Recombination) family of Markov chains [21] is a Markov chain Monte Carlo (MCMC) method that explores the space of valid redistricting plans by repeatedly merging two adjacent districts, sampling a random spanning tree of the merged subgraph, and cutting the tree to produce a new pair of districts. ReCom provides a principled sampling framework with strong empirical convergence evidence. However, ReCom was designed for political redistricting, i.e., it enforces population balance but not capacity constraints, operates on a single administrative level, and does not account for facility locations.

Our contribution. We introduce *FalCom* (Facility Location Combination), the first MCMC sampling framework for facility location and districting problems. The name reflects FalCom’s core operation: combining facility assignments with spatial partitions at multiple hierarchical levels. FalCom extends the spanning-tree partition paradigm of ReCom [21] in the following directions.

- (i) **Contiguity by construction.** Every district produced by FalCom is contiguous by construction: removing an edge from a spanning tree always yields two connected components. This distinguishes FalCom from the entire hierarchical facility location literature, where contiguity is almost universally absent.
- (ii) **Hierarchical capacitated partitioning.** FalCom operates on a multi-level graph structure with capacity constraints and demand balance at every level. The number of districts is not fixed a priori but emerges from the capacity requirements, making the framework adaptive to the problem instance.
- (iii) **Ensemble-based analysis.** Rather than seeking a single optimal solution, FalCom samples from the space of feasible hierarchical partitions, producing ensembles that enable distributional analyses: boundary frequency maps reveal which district boundaries are robust across the solution space, facility assignment stability identifies which facilities are essential versus substitutable, and capacity utilization profiles characterize the distribution of workloads. This ensemble perspective, pioneered in political redistricting for gerrymandering detection [21], has not previously been available for facility location or general districting under hierarchy and capacity constraints.
- (iv) **Scalability.** FalCom handles graphs with up to 50,000 basic units, at least one order of magnitude beyond most published districting and location-allocation methods. All operations (spanning tree sampling, feasibility scoring, and acceptance computation) are local to the merged subgraph and run in time linear in the subgraph size.
- (v) **Composability.** FalCom separates the combinatorially hard partition problem from within-district optimization. Flow-based formulations, MIP solvers, VRP heuristics, and stochastic models can all be applied *within* FalCom’s districts to optimize facility placement, route vehicles, or allocate staff. Conversely, any single solution produced by any method can be evaluated against the distributional baseline that FalCom provides.
- (vi) **Interactive visualizer.** We have released an open-source web-based visualizer that animates the FalCom chain step by step on the geographic graph, showing spanning tree construction, capacitated tree cuts, district assembly, and the accept/reject decision at each iteration. The tool requires no installation and can be used online [31].

The rest of the paper is organized as follows. Section 2 reviews the related literature. Section 3 introduces the problem formulation and the hierarchical graph structure, while Section 4 describes the state space and feasibility rules. In Section 5, we present the FalCom algorithm in detail, covering the proposal mechanism, recursive partitioning, capacitated tree cuts, and facility assignment. This is followed by Section 6, which discusses feasibility and the status of the convergence theory FalCom inherits from the recom family. Section 7 presents computational experiments on synthetic grids and a real-world case study emergency service zones in London. Finally, Section 8 provides conclusions and rich directions for future research.

2 Literature Overview

Our problem sits at the intersection of two largely separate literatures. The districting literature studies how to partition a graph or geographic region into balanced, contiguous zones, and has

recently produced Markov chain Monte Carlo methods that treat the partition itself as the state of a chain, but it does not assign capacitated facilities. The hierarchical facility-location literature, in turn, places capacitated facilities across multiple service tiers, but is overwhelmingly optimization-based, and the underlying districting structure is typically fixed or ignored. To our knowledge no prior work provides a tractable sampler over hierarchical capacitated partitions merging these two areas. We organize the related work along these two areas: Section 2.1 reviews districting problems, and Section 2.2 reviews hierarchical and capacitated facility location.

2.1 Districting Problems

The most comprehensive survey of districting problems classifies applications into four domains: political redistricting, sales territory design, service districting, and distribution districting, and catalogs the criteria (balance, contiguity, compactness) and formulations used in each [29]. Recent advances in optimal districting across multiple application areas are surveyed in [46]. Despite the diversity of applications, all four classes share a common combinatorial concept such as partitioning a planar graph into connected subgraphs.

Planning criteria. Three criteria recur across virtually all districting formulations [29].

Balance requires districts of equitable size with respect to one or more activity measures. The most common local measure defined on a district D_i is the relative deviation of the district demand $d(D_i)$ from the mean $\bar{d}$ over district demands, which is the constraint $|d(D_i) - \bar{d}| \leq \epsilon \bar{d}$ for a given tolerance ϵ [47, 51]. Some models balance multiple measures simultaneously, for example both the number of customers and the product demand per territory [48, 50].

Compactness asks that districts be somewhat round-shaped and undistorted. Distance-based measures are the most common: the weighted sum of distances from each unit to its district center [8, 25], the maximum pairwise distance within a district [47], or the Beardwood–Halton–Hammersley approximation to intra-district travel time. Optimizing compactness alone does not guarantee contiguity: squared distance measures can produce compact but disconnected districts [29].

Contiguity requires every district to induce a connected subgraph of the dual graph and is generally treated as a hard constraint [29]. Enforcing it is the dominant computational challenge in districting. Three mathematical formulations have been proposed. The separator (CUT) formulation writes, for each unit i assigned to center j and each i - j separator S , a constraint requiring at least one unit in S to also belong to district k . The number of such constraints is exponential, but they can be separated efficiently via minimum cuts, enabling a branch-and-cut approach [51, 54, 59]. The strongest results solve instances with up to 800–1,400 basic units and 12 districts to optimality [59]. The network-flow formulation models each district as a flow from its units to its center, requiring one unit of supply at each non-center node and conservation at intermediate nodes [54]. This uses $O(|V|^2)$ flow variables, which is polynomial but large. A simpler variant requires each district to be a subtree of a shortest-path tree rooted at its center [29]; this guarantees contiguity and compactness simultaneously but restricts the solution space. Heuristic enforcement checks contiguity during local search moves: adding a unit to a connected district preserves connectivity; removing a unit preserves it if the unit is not a cut vertex [45]. This is efficient, $O(|D_k|)$ per move where $|D_k|$ is the number of basic units in the district D_k , but limits the neighborhood structure and cannot certify global optimality. All three approaches incur substantial computational overhead. The ReCom framework [21] introduced a spanning-tree approach that guarantees contiguity by construction for political redistricting, but ReCom operates on a single administrative level and does not handle facility location or capacity constraints. FalCom extends this spanning-tree contiguity guarantee to general districting and hierarchical facility location.

Computational complexity. Districting problems are NP-hard in general. Even the basic problem of partitioning a planar graph into connected subgraphs of equal weight is strongly NP-hard [6]. Balanced graph partitioning, i.e., dividing a graph into k components with at most $(1 + \varepsilon)n/k$ nodes each while minimizing cut edges, is NP-hard for every fixed $k \geq 2$ and admits no constant-factor approximation unless $P=NP$ [5]. Adding contiguity constraints makes the problem harder. The separation problem for the exponential family of contiguity inequalities is itself NP-hard in the general case, though it becomes polynomial on planar graphs [59].

Applications. Within service districting, healthcare has emerged as a particularly active domain. Home-care districting in Montreal has been formulated under contiguity, borough-boundary, and workload-balance constraints, solved via tabu search [9]. Healthcare system partitioning in Paraná, Brazil, has been addressed using a multi-objective genetic algorithm [56]. A hierarchical healthcare districting model with robust optimization under demand uncertainty appears in [17], and a two-tier medical referral network design via hierarchical MILP with a hybrid genetic algorithm in [60]. Other application domains include police patrol sectors [11, 16], salt spreading and road maintenance [40], electrical power districting [8], meter reading [19], public transport [57], and last-mile delivery [52]. Stochastic extensions address demand uncertainty through multi-stage recourse models [43] and scenario-based robust formulations [17]. Table 1 summarizes the key attributes of representative models across domains, including modeling approach, objectives, solution methods, instance size, and whether hierarchical or capacitated structure is present.

Table 1: Implementations in the districting literature. Districting criteria: compactness, contiguity, and capacity–demand balance across districts. Facility criteria: hierarchy, capacity, location, and allocation. Implementation columns: real dataset validation, size of the largest instance, and the solution method.

Reference	Application	Districting criteria			Facility criteria				Implementation		
		Compact	Contig.	Balance	Hier.	Cap.	Location	Allocation	Real	Size	Method
2002 [40]	Salt spreading	✓	✓	✓					✓	500	Cluster-first heuristic
2002 [16]	Police districting	✓	✓	✓					✓	911	Simulated annealing
2003 [8]	Power distribution		✓	✓					✓	1,200	SA + local search
2003 [9]	Home care	✓	✓	✓	✓			✓	✓	82	Tabu search
2005 [30]	Territory design	✓	✓	✓					✓	400	MIP + local search
2007 [57]	Transportation	✓	✓						✓	1,300	ELECTRE TRI
2009 [47]	Commercial territory	✓	✓	✓					✓	5,000	GRASP
2010 [7]	Home care	✓	✓	✓						50	MIP
2011 [50]	Commercial design	✓	✓	✓					✓	5,000	GRASP
2012 [55]	Spatial allocation	✓	✓	✓						100	MIP
2013 [18]	Public administration	✓	✓	✓					✓	88	NSGA-II
2014 [19]	Power distribution		✓	✓					✓	2,000	MIP
2015 [11]	Police zones	✓	✓	✓					✓	500	MIP + local search
2015 [56]	Healthcare		✓	✓	✓			✓	✓	399	NSGA-II
2016 [28]	Graph partitioning		✓	✓						500	ACO
2019 [32]	Distribution			✓	✓			✓		100	MIP + heuristic
2020 [22]	Healthcare	✓	✓	✓	✓				✓	100	MIP
2020 [33]	Elderly care	✓	✓	✓	✓		✓	✓	✓	200	MINLP
2021 [17]	Healthcare		✓	✓	✓			✓	✓	100	Stochastic MIP
2021 [20]	Waste management		✓	✓					✓	399	NSGA-II
ReCom [21]	Political redistricting	✓	✓	✓	✓				✓	100,000	MCMC
2022 [52]	Last-mile delivery	✓	✓	✓	✓				✓	3,000	GRASP + ILS
2025 [60]	Healthcare network			✓	✓	✓	✓	✓	✓	200	Hybrid GA
FalCom	Districting framework	✓	✓	✓	✓	✓	✓	✓	✓	100,000	MCMC

Solution methods. The solution methods employed in the literature span a wide range. Location-allocation heuristics alternate between fixing facility locations and optimizing assignments [25]; the allocation phase can be solved optimally as a capacitated transportation problem for instances with 10,000 or more units when contiguity is relaxed [29]. Exact methods based on branch-and-cut with separator or flow-based contiguity constraints can solve instances with up to 800–1,400 basic units [51, 59]. Construction heuristics include multi-kernel growth [10] and successive dichotomies [30]. Among metaheuristics, GRASP has been particularly successful for commercial districting [47, 48], alongside tabu search [10], simulated annealing [16], and genetic algorithms [56, 57].

Districting without facilities. A notable feature of the districting literature is that facility location is typically treated as secondary. In home-care districting [9], for example, district centers are predetermined nurse bases, and the optimization focuses on balancing demands and workloads. In commercial territory design [47, 48], the goal is to partition customers into balanced territories, and the sales representative’s location is an afterthought. Capacity constraints, when present, bound the total weight per district rather than the throughput of a specific facility. To the best of our knowledge, no districting method in the literature jointly partitions a geographic region and assigns facilities with capacity at multiple hierarchical levels. This gap motivates the integration pursued in FalCom.

MCMC methods and ensemble analysis. The motivation for designing districting as a Markov chain Monte Carlo problem is ensemble analysis. Rather than producing a single optimum, the chain samples a large population of feasible partitions whose distributional properties, partisan-seat histograms, cut-edge counts, boundary-frequency maps, serve as the baseline against which any particular plan is judged. This paradigm originates in political redistricting, where ensemble statistics are the standard evidentiary tool for detecting partisan gerrymandering and have been used in U.S. federal court [21, 23]. The most relevant methodological predecessor of FalCom is the ReCom (Recombination) family of Markov chains [21]. At each step, ReCom samples a uniform spanning-tree and cuts edges from the tree, which makes the proposal kernel well defined, and contiguity holds by construction. On the other hand, the original construction is not reversible and detailed balance does not hold, i.e., ReCom samples from a stationary distribution that is well defined but not analytically characterized. Cannon et al. [12] address this with a reversible variant whose proposal preserves a known target distribution, it remains the most rigorous convergence guarantee available for spanning-tree-based redistricting chains. Mixing-time bounds for ReCom-type chains remain an open problem, and convergence is established in practice through empirical diagnostics such as Gelman–Rubin $\hat{R}$ and effective sample size across independent chains.

FalCom generalizes this line of work along hierarchical two-level proposals with a variable number of districts and capacitated facility assignment. The convergence theory FalCom needs and empirical diagnostics directly mirrors the trajectory of the redistricting community. Beyond politics, the ensemble paradigm becomes a tool for the broader districting and facility-location settings FalCom addresses with the questions which boundaries are robust across the feasible region, which facilities are essential versus substitutable, and how capacity utilization is distributed across plans, distributional questions that no single-optimum method can answer.

2.2 Hierarchical and Capacitated Facility Location

Hierarchical facility location problems (HFLPs) require locating facilities at multiple service levels and allocating customers to them across a hierarchy. The most widely used classification of HFLPs is due to Şahin and Süral [49], later extended by Contreras and Ortiz-Astorquiza [14] as follows.

Customers may request a single service type called *single demand* or several types simultaneously called *multiple demand*. When customers correspond to origin–destination pairs, demand takes the form of commodities that must be routed through the hierarchy. In a *successively inclusive* hierarchy, a level- ℓ facility provides all services offered at level-1 through $\ell - 1$ plus additional ones. In a *successively exclusive* hierarchy, each level offers only its own services. A *mixed* hierarchy has a non-empty but non-trivial intersection of service sets [42]. A *single-flow* pattern routes demand sequentially through every level of the hierarchy; a *multi-flow* pattern allows demand to skip levels or enter at any level; a *bidirectional-flow* pattern routes demand up and then back down. A system is *coherent* (or *single-assignment*) if all demand flowing into a lower-level facility is forwarded to exactly one higher-level facility, yielding a nested spatial structure; it is *non-coherent* if lower-level facilities may forward to several higher-level ones [49]. The three classical objective types are *fixed-charge* (minimize setup plus transportation cost), *median* (minimize demand-weighted distance for a given number of facilities), and *covering* (maximize the number of customers served within a coverage distance). FalCom is designed for the single-demand, successively exclusive, single-flow, coherent case, the simplest non-trivial hierarchical setting.

Following Contreras and Ortiz-Astorquiza [14], the literature organizes around four families. *Multi-level facility location problems* (MLFLPs) place facilities at every level under a successively-exclusive, single-flow hierarchy, with the path-based formulation of Tcha and Lee [58] as the canonical model. *Median and covering hierarchical location problems* (MHLPs and CHLPs) appear in healthcare, education, and emergency services, with the p - q -median variant [24, 53] and the maximum-coverage variant [38] as the canonical formulations and applications ranging from neonatal intensive care [13] to trauma networks [34]. *Multi-echelon location-routing problems* integrate vehicle routing with facility location at each echelon [15]; *hierarchical hub location problems* (HHLPs) route bidirectional origin–destination flows through multi-level hub-and-spoke networks [4, 62] and, unlike the other three families, typically do not require spatial contiguity of catchment areas. Across all four families, solution methods are dominated by mixed-integer programming and metaheuristics. No method we are aware of provides a Markov chain Monte Carlo sampler over the space of feasible hierarchical partitions.

The role of capacity. The intersection of hierarchy and capacity is especially challenging. Capacity constraints at each level couple the location and allocation decisions across levels, and the feasible region shrinks dramatically. In single-level problems, adding capacity to the uncapacitated FLP already makes the problem significantly harder [1]. In hierarchical settings, the coupling is tighter because the capacity consumed at one level constrains what is feasible at the next. Most hierarchical models in the literature are solved by MILP formulations that become intractable beyond a few hundred facilities and a few thousand demand points [14, 44]. Metaheuristic approaches extend tractability somewhat [17, 60], but even these methods scale to at most a few thousand demand nodes and do not produce distributional information about the solution space.

The contiguity gap. A striking feature of the entire HFLP literature is the near-total absence of spatial contiguity requirements. Classical formulations assign each demand node to the cheapest feasible facility regardless of geographic coherence, so that a single facility’s service area may be fragmented into disconnected districts [14]. For single-level problems, under a median objective the optimal districts are already compact and connected, because a disconnected district pays for its detached pocket through the long demand-weighted distance to its facility. Consequently much of the median-districting literature omits explicit contiguity constraints and recovers connected solutions for free. In hierarchical systems, contiguity is arguably even more important than in single-level

settings, for three reasons. First, referral chains are geographically coherent. A patient referred from a clinic to a hospital should not have to traverse another clinic’s territory, and fragmented service areas create confusing referral paths that increase both travel burden and administrative overhead. Second, administrative accountability requires that districts at each level correspond to recognizable geographic regions. Fragmented districts complicate staffing, management reporting, and public communication. Third, in logistics and distribution, non-contiguous service areas produce crossing delivery routes that increase vehicle-miles and operational costs [30, 40].

This distinction matters for FalCom. Every sampled state is contiguous by construction at every level, with no dependence on the objective for that guarantee rather than imposing contiguity through MIP constraints at all.

Gaps motivating FalCom. In summary, the HFLP literature exhibits four gaps that FalCom addresses simultaneously:

1. No existing method enforces contiguity as a structural guarantee in hierarchical settings. FalCom closes this gap by construction at zero computational cost.
2. No sampling framework explores the space of feasible hierarchical partitions.
3. Scalability beyond a few thousand demand points remains an open challenge for methods that enforce both hierarchy and capacity.
4. The number of districts is typically fixed a priori, rather than determined endogenously by capacity and demand.

Addressing these gaps starts with a precise statement of the problem the chain must respect, which we develop in the next section.

3 Problem Description and Model Formulation

We consider a hierarchical facility location framework with $L = \{1, \dots, h\}$ levels over a geographical region. The base graph $G^1 = (V^1, E^1)$ is a planar graph in which each node $i \in V^1$ corresponds to a basic geographic unit and each edge $ij \in E^1$ indicates that units i and j share a boundary. Every node $i \in V^1$ carries a demand value $d_i \geq 0$, and the total demand is $d = \sum_{i \in V^1} d_i$.

For each level $\ell \geq 2$, the graph $G^\ell = (V^\ell, E^\ell)$ is the *supergraph* of the partition at level $\ell - 1$, where each node in V^ℓ represents a district at level $\ell - 1$, and two nodes are adjacent in G^ℓ whenever their underlying districts share a boundary in G^1 . The demand of a node $i \in V^\ell$ is the aggregate demand of the base-level units it contains.

At each level ℓ , let $F^\ell \subseteq V^1$ denote the set of candidate nodes at which level- ℓ facilities may be located. The candidate sets are either pairwise disjoint or nested, i.e. either $F^\ell \cap F^{\ell'} = \emptyset$ for $\ell \neq \ell'$, or $F^{\ell+1} \subseteq F^\ell$, depending on the application. Since a basic unit is the smallest geographic entity, at most one facility per level may be placed in any given unit. Let $c_{\min}^\ell$ and $c_{\max}^\ell$ denote the minimum and maximum allowable capacity level respectively, w the maximum workload a service team can serve, and $\varepsilon^\ell \in (0, 1)$ the demand-balance tolerance at level ℓ . A facility at node j with capacity c_j^ℓ therefore supports a maximum workload of $w c_j^\ell$. When demand is concentrated, the model responds by increasing the capacity of existing or newly opened facilities rather than siting multiple facilities in the same unit, an assumption consistent with realistic operational constraints.

We require each service area to induce a connected subgraph of the base graph G^1 . We write this requirement as a closed-form Contiguity(x_{ij}^ℓ, G^ℓ) that returns 1 iff the set of base units $\{i \in V^1 : x_{ij}^\ell = 1\}$ assigned to facility j at level ℓ induces a connected subgraph of G^ℓ . We defer its concrete reformulation as a polynomial-size mixed-integer system to the paragraph after the constraint list, so the main formulation remains compact.

325 The decision variables are defined for each level $\ell \in L$. For each $j \in F^\ell$ and $i \in V^\ell$, let
326 $x_{ij}^\ell \in \{0, 1\}$ equal 1 if unit i is assigned to facility j at level ℓ , and 0 otherwise. Let $y_j^\ell \in \{0, 1\}$
327 equal 1 if facility j is open at level ℓ , and let $c_j^\ell \in \{0, 1, \dots, c_{\max}^\ell\}$ denote the capacity level assigned
328 to facility j . For clarity, all parameters and decision variables are summarized in the table below
329 before we state the model.

Table 2: Model parameters and sets.

Notation	Description
$L = \{1, \dots, h\}$	set of hierarchy levels
$G^\ell = (V^\ell, E^\ell)$	dual graph at level ℓ and G^1 is the base graph
V^ℓ	node set at level ℓ
E^ℓ	edge set at level ℓ ; adjacency inherited from G^1
$F^\ell \subseteq V^1$	candidate facility sites at level ℓ
d_i	demand of unit $i \in V^1$; aggregated for $i \in V^\ell$, $\ell \geq 2$
$d = \sum_{i \in V^1} d_i$	total demand
w	workload served by one service team level $\ell = 1$
$c_{\min}^\ell$	minimum capacity level per facility at level ℓ
$c_{\max}^\ell$	maximum capacity level per facility at level ℓ
ε^ℓ	demand-balance tolerance at level ℓ
b_j	cost of opening facility j
b	total budget

Table 3: Decision variables, defined for each level $\ell \in L$.

Variable	Description
$x_{ij}^\ell \in \{0, 1\}$	1 if unit $i \in V^\ell$ is assigned to facility $j \in F^\ell$
$y_j^\ell \in \{0, 1\}$	1 if facility $j \in F^\ell$ is open
$c_j^\ell \in \{0, 1, \dots, c_{\max}^\ell\}$	capacity level (number of service teams) at facility j

$$\sum_{j \in F^\ell} x_{ij}^\ell = 1, \quad \forall i \in V^1, \quad \forall \ell \in L \quad (1)$$

$$x_{ij}^\ell + x_{ij'}^{\ell-1} \leq 1 + x_{j'j}^\ell, \quad \forall i \in V^1, \quad j \in F^\ell, \quad j' \in F^{\ell-1}, \quad \ell \geq 2 \quad (2)$$

$$\text{Contiguity}(x_{ij}^\ell, G^\ell) = 1, \quad \forall j \in F^\ell, \quad \forall \ell \in L \quad (3)$$

$$c_{\min}^\ell y_j^\ell \leq c_j^\ell \leq c_{\max}^\ell y_j^\ell, \quad \forall j \in F^\ell, \quad \forall \ell \in L \quad (4)$$

$$\sum_{j \in F^\ell} y_j^\ell c_j^\ell \leq \lceil d/w \rceil, \quad \forall \ell \in L \quad (5)$$

$$(1 - \varepsilon^\ell) w c_j^\ell \leq \sum_{i \in V^1} d_i x_{ij}^\ell \leq (1 + \varepsilon^\ell) w c_j^\ell, \quad \forall j \in F^\ell, \quad \forall \ell \in L \quad (6)$$

$$x_{jj}^\ell = y_j^\ell, \quad \forall j \in F^\ell, \quad \forall \ell \in L \quad (7)$$

$$\sum_{j' \in F^1} z_{j'j} \geq \kappa_{\min}^2 y_j^2, \quad \text{where } z_{j'j} = y_{j'}^1 \wedge x_{j'j}^2, \quad \forall j \in F^2 \quad (8)$$

$$c_j^\ell \in [c_{\max}^\ell], \quad x_{ij}^\ell, y_j^\ell \in \{0, 1\}, \quad \forall i \in V^1, \quad j \in F^\ell, \quad \forall \ell \in L \quad (9)$$

The formulation captures the key structural elements of our problem. Constraint (1) ensures that every base unit $i \in V^1$ is assigned to exactly one facility at each level ℓ . Constraint (2) enforces hierarchical consistency. If base unit i is assigned to facility j at level $\ell \geq 2$ and to facility j' at level $\ell - 1$, then the base unit containing j' must itself be assigned to j at level ℓ . In other words, j' must lie within j 's service area, so that every level- ℓ district is a union of level- $(\ell-1)$ districts, which is a linking constraint for coherent hierarchical systems. Constraint (3) enforces contiguity of every service area at both levels by requiring the units assigned to facility j at level ℓ to induce a connected subgraph of G^1 . A specific contiguity formulation will be given below. The hierarchy constraint (2) forces every super-district to be a union of level-1 districts, but does not by itself force the super-district's footprint to be connected, so Contiguity is stated for every $\ell \in L$. Its mixed-integer reformulation is given in the next paragraph. Constraint (4) links facility opening to capacity at every level $\ell \in L$: a facility is open if and only if its capacity is at least $c_{\min}^\ell$, and its capacity cannot exceed $c_{\max}^\ell$.

Constraint (5) imposes an upper bound on the total number of service teams deployed at level ℓ . Since the total demand is d and each team can serve at most a workload of w , the aggregate capacity across all open facilities cannot exceed $\lceil d/w \rceil$. A lower bound on total capacity is not imposed because Constraint (6) already drives utilization toward equality by requiring the demand assigned to each facility j to fall within a $(1 \pm \varepsilon^\ell)$ tolerance of the maximum workload $w c_j^\ell$.

Constraint 7 forces open facilities to serve their own base units. This is implicit in Constraints (1)–(3) for every contiguous feasible plan, but stating it explicitly tightens the LP relaxation and removes a class of fractional optima that branch-and-cut would otherwise have to explore.

Constraint (9) imposes integrality and domain conditions on all decision variables.

Minimum number of level-1 districts per super-district (8). The parameter $\kappa_{\min}^2 \in \mathbb{Z}_{\geq 1}$ enforces that every open super-district at level 2 bundles at least $\kappa_{\min}^2$ open level-1 districts. The conjunction $z_{j'j} = y_{j'}^1 \wedge x_{j'j}^2$ is linearized in the standard way ($z \leq y_{j'}^1$, $z \leq x_{j'j}^2$, $z \geq y_{j'}^1 + x_{j'j}^2 - 1$).

Contiguity formulation. We realize the closed-form $\text{Contiguity}(x_{ij}^\ell, G^\ell)$ from (3) via the single-commodity (SHIR) flow reformulation of Shirabe [54], which replaces each occurrence of $\text{Contiguity}(x_{ij}^\ell, G^\ell) = 1$ by a polynomial-size linear system. The same flow argument applies at both levels because x_{ij}^ℓ assigns base units $i \in V^1$ to facility nodes $j \in F^\ell$ at *either* level ℓ . For each level $\ell \in L$, each facility $j \in F^\ell$, and each directed arc $(u, v) \in A^1$ (each undirected edge of G^1 traversed in both directions), introduce a non-negative flow variable $f_{uv}^{\ell,j}$. The facility node j acts as the source of one unit of flow per assigned node, and flow is admitted into a node only if that node is itself assigned to j at level ℓ :

$$f_{uv}^{\ell,j} \leq |V^1| x_{vj}^\ell, \quad \forall \ell \in L, j \in F^\ell, (u, v) \in A^1 \quad (10)$$

$$\sum_{(v,u) \in A^1} f_{vu}^{\ell,j} - \sum_{(u,v) \in A^1} f_{uv}^{\ell,j} = x_{uj}^\ell, \quad \forall \ell \in L, j \in F^\ell, u \in V^1 \setminus \{j\} \quad (11)$$

$$\sum_{(j,v) \in A^1} f_{jv}^{\ell,j} - \sum_{(v,j) \in A^1} f_{vj}^{\ell,j} = \sum_{u \in V^1} x_{uj}^\ell - x_{jj}^\ell, \quad \forall \ell \in L, j \in F^\ell \quad (12)$$

$$f_{uv}^{\ell,j} \geq 0, \quad \forall \ell \in L, j \in F^\ell, (u, v) \in A^1 \quad (13)$$

Constraints (10)–(13) are equivalent to $\text{Contiguity}(x_{ij}^\ell, G^1) = 1$ in the binary x^ℓ projection [59]: at each level ℓ they enforce contiguity of every service area assigned to j by requiring a flow path back to j through nodes that are themselves assigned to j at level ℓ . Both levels reuse the same base graph G^1 (super-districts inherit adjacency from the base graph through the dual graph G^2), so no separate supergraph flow is needed; level-2 contiguity becomes contiguity of the level-2 footprint on G^1 , which the SHIR system enforces directly via the $f^{2,j}$ variables. The total formulation adds $O((|F^1| + |F^2|) \cdot |E^1|)$ continuous variables and the same number of capacity constraints, plus $O((|F^1| + |F^2|) \cdot |V^1|)$ conservation constraints. SHIR is a self-contained polynomial-size model that requires no callbacks, and is what we use in the Gurobi benchmark of Section 7.3.

Absence of an objective function. The formulation above should be compared to classical capacitated facility location and districting models. The most important structural difference is the absence of an objective function. This is a design choice rooted in the sampling paradigm. In classical optimization, a single objective (e.g., minimize total demand-weighted distance) selects one solution from the feasible region. In practice, however, planners must balance multiple competing criteria such as travel time, workload equity, and facility utilization, whose relative weights are rarely known in advance. Rather than committing to a fixed weighting, FalCom treats the constraint set as defining the feasible region and samples from a distribution supported on it, biased by the candidate-awareness score ψ defined in Section 5.5. The resulting ensemble of feasible solutions enables distributional analyses (boundary frequency maps, facility stability profiles, capacity utilization distributions) that reveal which features of a plan are robust across the solution space and which are artifacts of a particular objective weighting. Extensions to explicit objective functions and additional constraints are discussed in Section 8.2.

Capacity as a decision variable. Unlike standard formulations in which facility capacities are fixed parameters, our model treats c_j^ℓ as a decision variable, simultaneously selecting facility locations and sizing their capacity. This is essential in settings where demand is heterogeneous and pre-specifying capacity would either waste resources or leave demand unmet.

Generality across districting domains. The formulation is intentionally generic: the four fundamental constraints: contiguity (3), demand balance (6), capacity (4), and hierarchical containment (2), appear in every districting domain, though the quantities they constrain change

meaning. In emergency service zone design, a district is a response zone, a facility is a station, capacity is the number of units stationed there, and demand balance equalizes call volume across zones. In commercial territory design, a district is a sales territory, a facility is a representative's base, capacity bounds the number of accounts a representative can manage, and balance equalizes revenue potential. In last-mile delivery, a district is a delivery zone, a facility is a micro-depot, and capacity is the number of parcels a courier can handle per shift. The same structural pattern extends to healthcare districting (catchment zones anchored by clinics), police zone design (patrol beats anchored by precinct stations), waste management (collection zones anchored by transfer stations), and power distribution (meter-reading districts anchored by substations).

4 Graph Formulation and Sample Space

The MIP formulation in Section 3 specifies what constitutes a valid solution. To move beyond finding a single optimal point, we reformulate the solution space as a combinatorial object on which a Markov chain can move, enabling us to *sample* configurations that are not merely feasible but collectively representative of the high-quality region.

4.1 Hierarchical State Space

The starting point is the *base dual graph* $G^1 = (V^1, E^1)$, introduced in Section 3: each node $i \in V^1$ represents a basic geographic unit and each edge $ij \in E^1$ indicates that units i and j share a boundary. A *partition* $\mathcal{P}^1 = \{D_1^1, D_2^1, \dots, D_{k_1}^1\}$ of G^1 splits the node set into pairwise disjoint districts, each inducing a connected subgraph of G^1 .

From a partition at any level, we construct a coarser graph one level up. The *supergraph* $G^2 = (V^2, E^2)$ of $\mathcal{P}^1$ has one node per district in $\mathcal{P}^1$ and one edge for each pair of adjacent districts. Like G^1 , the supergraph G^2 is itself a dual graph, where each node represents a contiguous region and each edge represents shared boundary, but of a coarser geographic partition. A partition $\mathcal{P}^2 = \{D_1^2, D_2^2, \dots, D_{k_2}^2\}$ of G^2 groups first-level districts into *superdistricts*, each inducing a connected subgraph of G^2 . Figure 1 illustrates the base dual graph and the supergraph arising from a partition.

The construction continues inductively: given the partition $\mathcal{P}^{\ell-1}$ at level $\ell - 1$, one builds the supergraph $G^\ell = (V^\ell, E^\ell)$ and partitions it to obtain $\mathcal{P}^\ell$. At every level ℓ , the graph G^ℓ is a dual graph of successively larger regions, and containment is ensured at every stage. The result is a nested family of partitions that refines from the bottom up.

We now formalize this construction. A *partition* $\mathcal{P}^\ell$ of G^ℓ at level $\ell \in L$ is

$$\mathcal{P}^\ell = \{D_1^\ell, D_2^\ell, \dots, D_{k_\ell}^\ell\} \quad \text{s.t.} \quad D_i^\ell \cap D_j^\ell = \emptyset, \quad \forall i \neq j, \quad (14)$$

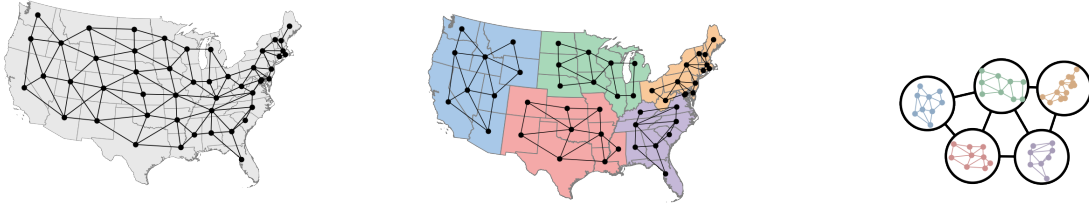
where each $D_i^\ell \subseteq V^\ell$ induces a connected subgraph of G^ℓ . An *h-hierarchical districting* is a family of partitions

$$\mathcal{P} = \{\mathcal{P}^1, \mathcal{P}^2, \dots, \mathcal{P}^h\} \quad \text{s.t.} \quad \bigcup_{i \leq k_\ell} D_i^\ell = \mathcal{P}^{\ell-1}, \quad \forall \ell \geq 2, \quad (15)$$

with the convention $\mathcal{P}^0 = V^1$. The containment condition in (15) ensures that every level- ℓ district is a union of level- $(\ell-1)$ districts, which is the graph-theoretic counterpart of the hierarchy constraint (2) in the MIP formulation.

At each level ℓ , the partition $\mathcal{P}^\ell$ is equipped with two functions. A *facility assignment*

$$f^\ell: \mathcal{P}^\ell \rightarrow F^\ell$$



(a) Geographic units with dual graph G . (b) A partition of G^1 into $k = 6$ contiguous districts. (c) Supergraph G^2 . Each supernode represents a district.

Figure 1: From geographic units to the supergraph used by FalCom.

maps each district to exactly one candidate facility site within it. A *capacity assignment*

$$c^\ell: \mathcal{P}^\ell \rightarrow \{1, 2, \dots, c_{\max}^\ell\}$$

specifies the number of service teams allocated to each district. In this paper, each district contains exactly one assigned facility, so $c^\ell(D_i^\ell)$ denotes both the district's capacity and that of its assigned facility. In other districting settings where multiple facilities may operate within a single district, $c^\ell(D_i^\ell)$ would represent the aggregate capacity of all assigned facilities in the district. For convenience, we extend the capacity notation to arbitrary subgraphs: for a subgraph H of G^ℓ , we write

$$c_H = \sum_{D_i^\ell \subseteq H} c^\ell(D_i^\ell)$$

for the *aggregate capacity* of H , i.e., the total number of service teams allocated to the districts contained in H . Each base-level node $v \in V^1$ carries a non-negative *demand* $d_v \geq 0$, as introduced in Section 3; we use $d(v) \equiv d_v$ interchangeably and reserve the two-argument form $d(\cdot, \cdot)$ for the travel distance metric used in Section 5.5. For higher levels $\ell \geq 2$, the demand of a supernode $v \in V^\ell$ is the aggregate demand of the base-level units it contains. Extending this to subgraphs, for any subgraph H of G^ℓ we write

$$d_H = \sum_{v \in G^1[H]} d_v$$

for the *total demand* of the base-level units contained in H . When H is a subgraph of the base graph G^1 , this reduces to $d_H = \sum_{v \in V(H)} d_v$. For individual districts and subtrees, we use function notation $d(D_i^\ell)$ and d_H to distinguish partition elements from graphs.

We emphasize that the number of districts $k_\ell = |\mathcal{P}^\ell|$ is not fixed in advance: it is determined jointly by the capacity constraint and the demand distribution. Two solutions in Ω may have different numbers of districts at the same level.

A *solution* to the hierarchical capacitated facility location problem is a tuple

$$s = (\mathcal{P}^\ell, f^\ell, c^\ell \mid \ell \in L) \in \Omega, \quad (16)$$

where Ω denotes the set of all such tuples.

4.2 Constraint Rules and Feasibility

The graph-partitioning framework automatically satisfies several constraints from the MIP formulation in Section 3:

- **Unique assignment** (1): since $\mathcal{P}^\ell$ is a partition, every unit belongs to exactly one district.
- **Hierarchical containment** (2): enforced by the nesting condition (15).
- **Facility-capacity link** (4) Opening decision of a facility is linked to its capacity level by the definition of the assignment function f^ℓ .
- **Integrality** (9): the functions f^ℓ and c^ℓ are integer-valued by construction.

The remaining constraints require explicit enforcement. We collect them into a rule set $\mathcal{C} = \bigcup_{\ell \in L} \mathcal{C}^\ell$, where

$$\mathcal{C}^\ell = \{\text{contiguity}(\mathcal{P}^\ell), \text{balance}(\mathcal{P}^\ell, c^\ell), \text{coverage}(\mathcal{P}^\ell, c^\ell)\}. \quad (17)$$

Each rule corresponds to a MIP constraint from Section 3: *contiguity* requires that every $G^\ell[D_i^\ell]$ is connected, corresponding to constraint (3); *balance* requires $(1-\varepsilon^\ell) w c^\ell(D_i^\ell) \leq d(D_i^\ell) \leq (1+\varepsilon^\ell) w c^\ell(D_i^\ell)$ for each district, corresponding to constraint (6); *coverage* bounds the aggregate capacity, corresponding to constraint (5).

In FalCom (Section 5), each rule is satisfied by a specific algorithmic phase: contiguity is guaranteed by the spanning tree structure (Section 5.4), balance is enforced by the feasibility score ϕ during tree cuts (Section 5.4), and coverage is controlled by the recursive partitioning phase (Section 5.3).

A solution $s \in \Omega$ is *feasible* if it satisfies every rule in $\mathcal{C}$. The *feasible state space* is

$$\Omega_{\mathcal{C}} = \{s \in \Omega : s \text{ satisfies all constraints in } \mathcal{C}\}. \quad (18)$$

Each $s \in \Omega_{\mathcal{C}}$ is a *state* of FalCom.

The correspondence to the MIP of Section 3 is direct: the partition $\mathcal{P}^\ell$ encodes the assignment variables x_{ij}^ℓ , the facility assignment f^ℓ encodes the opening variables y_j^ℓ , and the capacity function c^ℓ encodes c_j^ℓ . Finding a feasible MIP solution $(x_{ij}^\ell, y_j^\ell, c_j^\ell)$ satisfying constraints (1)–(9) is equivalent to finding a state $s \in \Omega_{\mathcal{C}}$.

5 FalCom: A Sampling Method

FalCom extends the ReCom (Recombination) framework of DeFord et al. [21] from political redistricting to hierarchical capacitated facility location. ReCom explores the space of valid partitions by repeatedly merging two adjacent districts, sampling a uniform random spanning tree of the merged subgraph, and cutting the tree along a balanced edge to produce two new districts. Contiguity is guaranteed by construction, since removing an edge from a spanning tree yields two connected components. However, ReCom operates on a single administrative level, enforces only population balance, and is blind to facility locations. FalCom generalizes this paradigm in three directions: (1) a hierarchical two-level proposal that operates on a supergraph to localize computation, (2) a recursive partitioning scheme that produces a variable number of districts under capacity constraints, and (3) a candidate-awareness score that biases cut selection toward districts with centrally located facilities.

5.1 Outline of FalCom

The design of FalCom rests on three ideas. First, partitioning a graph by sampling a spanning tree and cutting an edge produces connected components by construction, contiguity is never

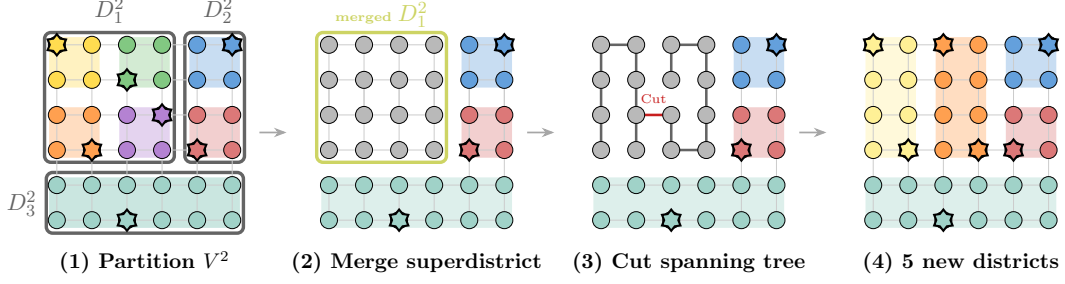


Figure 2: One iteration of FalCom: (1) Partition supergraph; (2) Merge a superdistrict; (3) a spanning tree is built on the merged region and a feasible cut edge is sampled; (4) the resulting new state s_{i+1} . Stars indicates candidate facilities.

violated and never needs to be repaired. Second, by operating on the supergraph G^2 to select *which region* of the base graph to re-partition, each iteration touches only a local subgraph rather than the full graph, giving per-step cost proportional to the merged region rather than $|V^1|$. Third, a candidate-awareness score biases cut selection toward districts with centrally located facilities, steering the stationary distribution toward high-quality configurations.

FalCom consists of four phases: a *hierarchical proposal* phase (Section 5.2), a *recursive partitioning* phase (Section 5.3), a *capacitated tree cut* phase (Section 5.4), and a *facility assignment* phase (Section 5.5). We present the method for two levels ($|L| = 2$); the extension to $|L| > 2$ is immediate by induction on the levels of hierarchy.

We write $G^2(\mathcal{P}^1)$ for the supergraph induced by a partition $\mathcal{P}^1$, as defined in Section 4.1. The chain is initialized with a feasible partition $s^{(0)} \in \Omega_C$ produced by applying the recursive partitioning phase to the full base graph (see the initialization paragraph after Algorithm 1). Each iteration (i) produces a candidate next state for iteration ($i + 1$)

$$s^{(i)} = (\mathcal{P}^\ell, f^\ell, c^\ell \mid \ell = 1, 2)^{(i)} \mapsto s^{(i+1)} = (\mathcal{P}^\ell, f^\ell, c^\ell \mid \ell = 1, 2)^{(i+1)}$$

via the following inter-level strategy:

1. **Upper level.** Build the supergraph $G^2((\mathcal{P}^1)^{(i)})$ and partition it into new superdistricts $(\mathcal{P}^2)^{(i+1)}$ with capacity assignment $(c^2)^{(i+1)}$.
2. **Lower level.** Choose one superdistrict $D^2 \in (\mathcal{P}^2)^{(i+1)}$ uniformly at random. Re-partition the induced subgraph $G^1[D^2]$ into new districts $D_1^1, \dots, D_m^1$ with capacity assignment $(c^1)^{(i+1)}$, while keeping all districts outside D^2 unchanged.
3. **Facility assignment.** Assign $(f^1)^{(i+1)}$ and $(f^2)^{(i+1)}$ independently of the current state $s^{(i)}$.
4. **Accept/reject.** If candidate-awareness score defined in Section 5.5 has $\psi > 0$ for all cuts used in the proposal, accept $s^{(i+1)} \leftarrow s'$; otherwise reject and set $s^{(i+1)} \leftarrow s^{(i)}$.

Steps 1 and 2 together constitute the *hierarchical proposal phase* (Section 5.2), which orchestrates the inter-level interaction: it constructs the input tuple $(H, c_H, \bar{d}_H, \varepsilon^\ell)$ that the recursive partitioning phase requires, and chooses the demand target that anchors local balance. Each step then invokes the recursive partitioning phase (Section 5.3), which in turn calls the capacitated tree cut phase (Section 5.4) as a subroutine. In step 1, solving for $(\mathcal{P}^2)^{(i+1)}, (c^2)^{(i+1)}$ amounts to partitioning the supergraph $G^2((\mathcal{P}^1)^{(i)})$ with a feasible capacity assignment, using the global average demand per team $\bar{d}$. In step 2, the same phase is invoked on the induced subgraph $G^1[D^2]$ of the selected superdistrict with a *local* demand target $\bar{d}_H = d_H/c_H$, which Section 5.2 motivates as necessary whenever supergraph density deviates from the global average. Step 3 assigns facilities fresh at each

iteration, independently of $s^{(i)}$, which is crucial for the mixing time of the chain (see Section 5.5). Figure 2 illustrates one iteration of the inter-level strategy.

Acceptance rule. After constructing the candidate state s' , FalCom accepts or rejects it. All rules in $\mathcal{C}$ are satisfied by construction during the proposal phases: contiguity by the spanning tree structure, capacity-demand balance by the feasibility score ϕ defined in Section 5.4, and total capacity-demand coverage by the recursive partitioning phase. A proposal is accepted if all phases complete successfully; otherwise, the chain remains at the current state $s^{(i)}$ and a new proposal is generated in the next iteration.

Algorithm 1 summarizes one iteration.

Algorithm 1: Outline of FalCom for $|L| = 2$

Input: Graphs G^1, G^2 ; current state $s^{(i)} = (\mathcal{P}^\ell, f^\ell, c^\ell \mid \ell = 1, 2)^{(i)}$; parameters $\varepsilon, w, c_{\max}$

Output: Next state $s^{(i+1)}$

```

1 Partition  $G^2$  into superdistricts  $(\mathcal{P}^2)^{(i+1)}$  with  $(c^2)^{(i+1)}$ ;           // Sections 5.3, 5.4
2 Assign facilities  $(f^2)^{(i+1)}: (\mathcal{P}^2)^{(i+1)} \rightarrow F^2$ ;                 // Section 5.5
3 Choose superdistrict  $D^2 \in (\mathcal{P}^2)^{(i+1)}$  uniformly at random;
4  $\mathcal{P} \leftarrow (\mathcal{P}^1)^{(i)} \setminus \{D \in (\mathcal{P}^1)^{(i)} : D \subseteq D^2\}$ ;           // preserve unchanged districts
5 Partition  $G^1[D^2]$  into districts  $D_1^1, \dots, D_m^1$  with  $(c^1)^{(i+1)}$ ;     // Sections 5.3, 5.4
6  $(\mathcal{P}^1)^{(i+1)} \leftarrow \mathcal{P} \cup \{D_1^1, \dots, D_m^1\}$ ;
7 Assign facilities  $(f^1)^{(i+1)}: (\mathcal{P}^1)^{(i+1)} \rightarrow F^1$ ;                 // Section 5.5
8  $s' \leftarrow (\mathcal{P}^\ell, f^\ell, c^\ell \mid \ell = 1, 2)^{(i+1)}$ ;
9 if  $\psi > 0$  for all cuts used in the proposal;                             // Section 5.4
10 then
11    $s^{(i+1)} \leftarrow s'$ ;
12 else
13    $s^{(i+1)} \leftarrow s^{(i)}$ ;                                           // reject: stay at current state
14 end
15 return  $s^{(i+1)}$ 

```

Initialization. Our implementation therefore constructs a feasible initial state by applying the recursive partitioning phase described in Section 5.3 directly to the full base graph G^1 . Given the total demand d_{G^1} and the maximum per-team workload w , the number of teams or the total capacity c_G equivalently is set to $c_G = \lfloor d_{G^1}/w \rfloor$, and Algorithm 3 is called with the inputs $(G^1, c_{G^1}, \bar{d}, \varepsilon^1)$ to recursively partition the graph G^1 yielding an initial partition $(\mathcal{P}^1)^{(0)}$ together with a capacity assignment $(c^1)^{(0)}$ for a given ε^1 . The supergraph $G^2((\mathcal{P}^1)^{(0)})$ is then built, and a second call to *RecursivePartitioning* on G^2 yields the initial superdistrict partition $(\mathcal{P}^2)^{(0)}$ with $(c^2)^{(0)}$. Facility assignments are computed deterministically via Equation (31). The resulting initial state $s^{(0)} \in \Omega_{\mathcal{C}}$ is feasible by construction, and the chain begins producing valid samples immediately.

5.2 Hierarchical Proposal Phase

As outlined in Section 5.1, the supergraph is partitioned at every state of the chain, and FalCom re-designs the districts within one of the resulting superdistricts. Both processes are carried out by the recursive partitioning phase in Section 5.3. The hierarchical proposal phase manages the

interaction between the two levels. At every state, it constructs the input tuple that the recursive partitioning phase requires and ensures that the capacity assignments are consistent across levels.

The recursive partitioning phase takes the input $(H, c_H, \bar{d}_H, \varepsilon^\ell)$, where H is the connected subgraph to be partitioned, c_H is the aggregate capacity, where each capacity corresponds to a service team to allocate, $\bar{d}_H$ is the local demand target per team, and ε^ℓ is the balance tolerance at level ℓ . The hierarchical proposal phase constructs this tuple differently for the upper and lower levels. Let $\bar{d} = d_{G^1}/t$ denote the *global* average demand per team, where $t = \lfloor d_{G^1}/w \rfloor$ is the total number of service teams.

Upper level: Solve for $(\mathcal{P}^2)^{(i+1)}$, $(c^2)^{(i+1)}$ using $(\mathcal{P}^1)^{(i)}$. We first construct the supergraph $G^2((\mathcal{P}^1)^{(i)})$. The total capacity at level 2 equals the sum of level-1 capacities, i.e., $c_{G^2} = \sum_{D_i^1 \in \mathcal{P}^1} c^1(D_i^1)$. We pass the input tuple $(G^2, c_{G^2}, \bar{d}, \varepsilon^2)$ to the recursive partitioning phase, which partitions the supergraph into new superdistricts $(\mathcal{P}^2)^{(i+1)} = \{D_1^2, D_2^2, \dots, D_k^2\}$ with capacity assignment $(c^2)^{(i+1)}$. Here $\bar{d}$ is the global average: since the supergraph represents the entire region, no local adjustment is needed.

Lower level: Solve for $(\mathcal{P}^1)^{(i+1)}$, $(c^1)^{(i+1)}$ using $(\mathcal{P}^2)^{(i+1)}$. After the upper level is settled, one superdistrict $D^2 \in (\mathcal{P}^2)^{(i+1)}$ is selected uniformly at random. Let $H = G^1[D^2]$ be the subgraph of G^1 induced by the base-level units in this superdistrict. The total capacity to allocate in H is $c_H = c^2(D^2)$, and we compute a *local* average demand per team $\bar{d}_H = d_H/c_H$. We then pass the input tuple $(H, c_H, \bar{d}_H, \varepsilon^1)$ to the recursive partitioning phase.

A local average is necessary because the demand density within the selected superdistrict may differ substantially from the global density; using the global target $\bar{d}$ would systematically over- or under-allocate teams in dense or sparse regions, which is explained in more detail in Section 5.3.

Algorithm 2 summarizes the hierarchical proposal phase.

Algorithm 2: Hierarchical proposal

Input: G^1, G^2

Output: $\mathcal{P}^1, \mathcal{P}^2, c^1, c^2$

// Upper level: supergraph partition

1 $c_{G^2} \leftarrow \sum_{D_i^1 \in \mathcal{P}^1} c^1(D_i^1)$;

2 $\mathcal{P}^2, c^2 \leftarrow \text{RecursivePartitioning}(G^2, c_{G^2}, \bar{d}, \varepsilon^2)$; // Section 5.3

3 Select superdistrict $D^2 \in \mathcal{P}^2$ uniformly at random;

// Lower level: re-split selected superdistrict

4 $H \leftarrow G^1[D^2]$;

5 $\bar{d}_H \leftarrow d_H / c_H$;

6 $\mathcal{P}^1, c^1 \leftarrow \text{RecursivePartitioning}(H, c_H, \bar{d}_H, \varepsilon^1)$; // Section 5.3

7 **return** $\mathcal{P}^1, \mathcal{P}^2, c^1, c^2$

The key insight is that $\bar{d}_H$ is recomputed locally for the selected superdistrict at the lower level. This allows the lower-level split to adapt to the demand density of the specific region being re-partitioned, rather than using the global workload target $\bar{d}$.

The two-task structure above is one specific inter-level design: the upper level is re-partitioned at every iteration, and the lower level adapts its workload target to the resulting superdistricts. Other designs are possible depending on the application. For instance, in problems where upper-level boundaries are administratively fixed (e.g., existing precinct boundaries in police zone design), the supergraph partition can be held constant and only the lower-level re-split is performed. Similarly,

the capacity transfer rule, how superdistrict capacity $c^2(D^2)$ constrains the lower-level split, can be relaxed or tightened depending on whether the application requires strict nesting or allows flexibility across levels. We adopt the fully dynamic design because it provides the most general exploration of the state space; a systematic comparison of alternative inter-level designs is deferred to a follow-up study.

5.3 Recursive Partitioning Phase

The recursive partitioning phase takes the input $(H, c_H, \bar{d}_H, \varepsilon^\ell)$ constructed by the hierarchical proposal phase in Section 5.2, and divides H into a collection of contiguous, capacity-demand balanced connected subgraphs of H , each carrying a capacity level and containing at least one facility candidate. The phase is called by both tasks of the hierarchical proposal, once on the supergraph $H = G^2$ to produce superdistricts, and once on a base-level subgraph $H \subset G^1$ to produce districts.

The phase works by repeatedly extracting one district at a time from a shrinking residual graph, using the capacitated tree cut phase detailed in Section 5.4, until the entire graph is partitioned. Each extracted district D_r must satisfy the *global balance rule* $\text{balance}(\mathcal{P}^\ell, c^\ell) \in \mathcal{C}^\ell$ given by

$$d(D_r) \in I := [(1 - \varepsilon^\ell)c^\ell(D_r)\bar{d}, (1 + \varepsilon^\ell)c^\ell(D_r)\bar{d}] \quad (19)$$

$$= [c^\ell(D_r)\bar{d} - \tau^\ell, c^\ell(D_r)\bar{d} + \tau^\ell] \quad (20)$$

defined in Section 4.2, where $\tau^\ell = \varepsilon^\ell c^\ell(D_r)\bar{d}$ is called *global tolerance*. The *global coverage rule* $\text{coverage}(\mathcal{P}^\ell, c^\ell) \in \mathcal{C}^\ell$ requires a *local coverage rule*

$$\sum_{r=1}^k c^\ell(D_r) = c_H. \quad (21)$$

The sequential extraction produces districts whose demands deviate from $\bar{d}_H$, and these deviations accumulate during the recursion. To ensure that the cumulative deviation does not violate the global balance and coverage rules for later districts, we design a debt-tracking mechanism and define local constraints accordingly. Without correction, the residual at the end of the recursion may carry a leftover demand that cannot simultaneously absorb the remaining capacity and respect (19).

Recursion and residual graph. Let $H^{(0)} := H$ be the initial graph of the recursion at the hierarchical level $\ell = 1, 2$. At each step $r \geq 1$, we extract a district D_r from the previous residual graph $H^{(r-1)}$, yielding the current residual

$$H^{(r)} := H^{(r-1)} \setminus D_r,$$

where $H^{(r-1)} \setminus D_r$ denotes the subgraph of $H^{(r-1)}$ induced by $V_{H^{(r-1)}} \setminus D_r$. The process terminates after k recursion steps when the remaining capacity reaches zero, producing the sequence of residual graphs

$$H^{(0)} \supset H^{(1)} \supset \dots \supset H^{(k)} = \emptyset,$$

where inclusion denotes induced subgraph containment. The resulting partition of V_H is $\mathcal{P}_H^\ell = \{D_1, \dots, D_k\}$, where $V_H = \bigsqcup_{r=1}^k D_r$ with $D_r \neq \emptyset$ for all r , and each D_r induces a connected subgraph of $H^{(r)}$. A capacity level $c^\ell(D_r) \in [c_{\min}^\ell, c_{\max}^\ell]$ is simultaneously assigned to each district D_r during the extraction process, updating the remaining capacity $c^{(r)}$ for the residual graph $H^{(r)}$ by

$$c_{H^{(0)}} = c_H, \quad c_{H^{(r)}} = c_{H^{(r-1)}} - c^\ell(D_r). \quad (22)$$

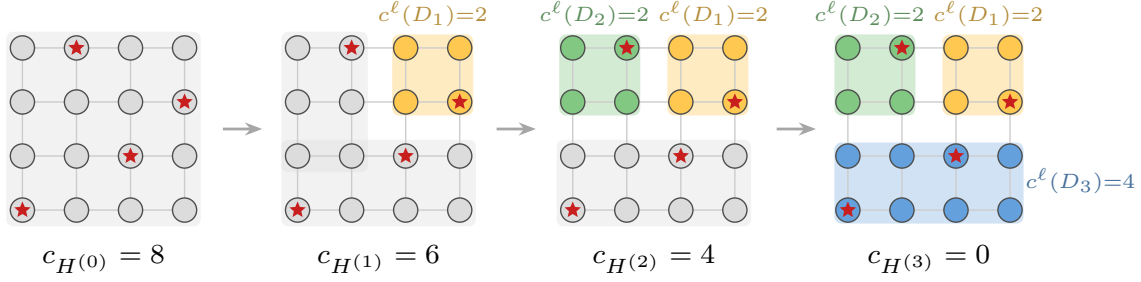


Figure 3: Recursive partitioning phase with an initial graph $H^{(0)}$ and teams $c_{H^{(0)}} = 8$. Districts D_1, D_2, D_3 are extracted sequentially from the initial graphs $H^{(0)}, H^{(1)}, H^{(2)}$, resulting remaining teams 6, 4, 0 service teams, respectively. Stars mark facility candidates.

as shown in Figure 3. Since each step consumes at least $c_{\min}^\ell \geq 1$ units of capacity, the recursion terminates in $k \leq c_H$ steps, when $c_{H^{(r)}}$ reaches zero, and the local coverage rule (21) holds by construction. The number of districts k is therefore determined endogenously by the recursion, not fixed a priori. We next introduce the debt mechanism and show how we control demand deviations to preserve the global balance rule (19).

Debt-track mechanism. When we extract the first district D_1 from $H^{(0)}$, the global balance rule (19) permits $d(D_1) \neq c^\ell(D_1) \bar{d}$, so the residual $H^{(1)} := H^{(0)} \setminus D_1$ carries a slightly skewed demand-to-capacity ratio. Sequential extraction accumulates such deviations and, left uncorrected, can leave an end-of-recursion residual whose demand cannot satisfy the remaining capacity assignment within tolerance. To compensate, we track the cumulative deviation via a *demand debt*

$$\delta^{(0)} = 0, \quad \delta^{(r)} = \delta^{(r-1)} + (d(D_r) - c^\ell(D_r) \bar{d}), \quad (23)$$

updated after extracting D_r with capacity $c^\ell(D_r)$. We measure global balance *per team*: with the per-team tolerance $\tau^\ell := \varepsilon^\ell \bar{d}$, every extracted district must satisfy

$$\frac{d(D_r)}{c^\ell(D_r)} \in J^\ell := [\bar{d} - \tau^\ell, \bar{d} + \tau^\ell] = \bar{d} [1 - \varepsilon^\ell, 1 + \varepsilon^\ell], \quad (24)$$

so the capacity-indexed district window scales linearly with c , $I^\ell(c) := c \cdot J^\ell = [c\bar{d} - c\tau^\ell, c\bar{d} + c\tau^\ell]$, and the global balance rule $d(D_r) \in I^\ell(c^\ell(D_r))$ is the per-team condition multiplied by $c^\ell(D_r)$.

The debt is used to asymmetrically shift the per-team window. For each candidate capacity $c \in [c_{\min}^\ell, c_{\max}^\ell]$ with $c \leq c_{H^{(r-1)}}$, we define the *debt-corrected per-team window* at recursion step r by

$$\tilde{L}^{(r)} := \max\{\bar{d} - \tau^\ell, \bar{d} - \tau^\ell - \delta^{(r-1)}\}, \quad (25)$$

$$\tilde{U}^{(r)} := \min\{\bar{d} + \tau^\ell, \bar{d} + \tau^\ell - \delta^{(r-1)}\}, \quad (26)$$

and clip the candidate at capacity c to $I^{(r)}(c) := c \cdot [\tilde{L}^{(r)}, \tilde{U}^{(r)}]$. The constraint $c \leq c_{H^{(r-1)}}$ prevents allocating more than the remaining capacity. When $\delta^{(r-1)} > 0$ earlier extractions overshot $\bar{d}$ in aggregate, so the min in (26) pulls the upper bound down by $\delta^{(r-1)}$ while the max in (25) leaves the lower bound at $\bar{d} - \tau^\ell$. The symmetric statement applies when $\delta^{(r-1)} < 0$. The clipping max/min guarantees $[\tilde{L}^{(r)}, \tilde{U}^{(r)}] \subseteq J^\ell$, hence $I^{(r)}(c) \subseteq I^\ell(c)$ for every c : any district with $d(D_r) \in I^{(r)}(c)$ automatically satisfies the global balance rule (19). The capacitated tree-cut phase in Section 5.4 seeks a connected $D \subseteq V_{H^{(r-1)}}$ and a capacity c with $d(D) \in I^{(r)}(c)$. We then set $D_r = D$ and

661 $c^\ell(D_r) = c$. Obtaining $I^{(r)}(c)$ as large as possible is better for feasibility; alternative choices of τ^ℓ
 662 and window shape are discussed in Appendix A. Theorem A.1 shows that $[\tilde{L}^{(r)}, \tilde{U}^{(r)}]$ has width at
 663 least τ^ℓ at every step, and that the invariant $|\delta^{(r)}| \leq c_{\max}^\ell \tau^\ell$ is preserved throughout the recursion.

664 **Computational cost and safety check.** The capacitated tree-cut routine (Section 5.4) re-
 665 samples a spanning tree whenever the current tree admits no cut in the debt-corrected window $I^{(r)}(c)$,
 666 up to a fixed budget of M attempts. Each attempt costs $O(|V(H^{(r-1)})|)$ for Wilson's spanning-tree
 667 sampler plus $O(|V(H^{(r-1)})|)$ to enumerate the at most $|V(H^{(r-1)})| - 1$ tree-induced cuts. If all M
 668 attempts fail, the entire proposal is rejected and the chain stays at $s^{(i)}$, so the worst-case per-step
 669 cost is $O(M \cdot |V(H^{(r-1)})|)$ for the rare unsuccessful proposals and $O(|V(H^{(r-1)})|)$ in the typical
 670 case. As a safety net we additionally check the per-team balance (24) on the returned cut before
 671 committing it; the clipping $[\tilde{L}^{(r)}, \tilde{U}^{(r)}] \subseteq J^\ell$ in (25)–(26) makes this check unreachable under the
 672 debt-corrected window, but should a numerical edge case violate it, the proposal is rejected and the
 673 chain remains at $s^{(i)}$. No additional re-calculation is triggered by a safety-net rejection.

674 We summarize the recursive partitioning phase via Algorithm 3 below.

Algorithm 3: Recursive partitioning phase at level ℓ

Input: Subgraph H , capacity c_H , demand target $\bar{d}_H$, tolerance ε^ℓ , capacity limit $c_{\max}^\ell$

Output: Partition $\mathcal{P}_H$, capacity assignment c^ℓ

```

1   $r \leftarrow 1$ ;
2   $\delta^{(0)} \leftarrow 0$ ;
3   $H^{(0)} \leftarrow H$ ;
4   $c_{H^{(0)}} \leftarrow c_H$ ;
5  while  $c_{H^{(r-1)}} > 0$  do
6      Compute  $I^{(r)} = \{I^{(r)}(c) \mid c \in [c_{\min}^\ell, c_{\max}^\ell], c \leq c_{H^{(r-1)}}\}$ ; // Equations (25)–(26)
7       $D_r, c^\ell(D_r) \leftarrow \text{CapacitatedTreeCut}(H^{(r-1)}, c_{H^{(r-1)}}, I^{(r)}, c_{\max}^\ell)$ ; // Section 5.4
8      Compute  $\delta^{(r)}$ ; // Equation (23)
9       $H^{(r)} \leftarrow H^{(r-1)} \setminus D_r$ ;
10      $c_{H^{(r)}} = c_{H^{(r-1)}} - c^\ell(D_r)$ ;
11      $r \leftarrow r + 1$ 
12 end
13 return  $\mathcal{P}_H = \{D_1, \dots, D_k\}$  and  $c^\ell$ 

```

676 The core operation in each iteration of Algorithm 3 is the call to *CapacitatedTreeCut* (line 7),
 677 which extracts a single district D_r with its capacity assignment $c^\ell(D_r)$ from the residual graph
 678 $H^{(r-1)}$. In fact, capacitated tree cut is the core operation of the whole FalCom. We now describe
 679 this subroutine in detail.

680 5.4 Capacitated Tree Cut Phase

681 The capacitated tree cut phase performs the actual district extraction at each step of the recursive
 682 partitioning. Given the input $(H^{(r-1)}, c_{H^{(r-1)}}, I^{(r)}, c_{\max}^\ell)$ at each recursion step r of an hierarchical
 683 level $\ell = 1, 2$, it samples a random spanning tree $T^{(r-1)}$ of the residual graph $H^{(r-1)}$ by Wilson's
 684 algorithm [61], and identifies the subtrees whose extraction would yield an admissible district under
 685 the conditions described below, and selects one such subtree with probability proportional to its

candidate-awareness score introduced in Section. 5.5. The selected subtree's node set becomes the extracted district D_r with a capacity $c^\ell(D_r) \in [c_{\min}^\ell, c_{\max}^\ell]$. The admissible cuts are determined by Algorithm 5, and the overall extraction is performed by Algorithm 4. Both algorithms operate unchanged at the base level $\ell = 1$ and the supergraph level $\ell = 2$, only the admissibility conditions differs.

Spanning tree cuts. Let $T^{(r-1)}$ be a spanning tree of $H^{(r-1)}$ rooted at an arbitrary node v^* . For any node u , let T_u denote the subtree rooted at u and $T_{u'} := T^{(r-1)} \setminus T_u$ its complement. When $u \neq v^*$, removing the edge uu' splits $T^{(r-1)}$ into T_u and $T_{u'}$. The case $u = v^*$ corresponds to $T_u = T^{(r-1)}$ and $T_{u'} = \emptyset$, which cannot be expressed by removing any edge. We therefore phrase the algorithm in terms of subtrees rather than edge cuts. We describe the admissibility condition for a subtree T_u at both levels $\ell = 1, 2$ below, and treat the node set $V(T_u)$ as the candidate district to be extracted. Removing an edge from a spanning tree always yields two connected components, so the rule contiguity($\mathcal{P}^\ell$) $\in \mathcal{C}^\ell$ is guaranteed by construction. A single depth-first traversal of $T^{(r-1)}$ from v^* computes the demand $d(T_u)$ of every subtree T_u in $O(|H^{(r-1)}|)$ total time, so the per-step cost of FalCom is linear in the residual graph size.

Feasibility function. The two levels share the same control structure but differ in what counts as an admissible subtree. At the base level $\ell = 1$, both T_u and its complement $T_{u'}$ must contain at least one candidate facility from F^1 , one will be the extracted district, and the other becomes the residual that must itself be partitionable in the following steps. At the supergraph level $\ell = 2$, no facility constraint is imposed, but a size constraint $\kappa_{\min}^2 \in \mathbb{Z}_{\geq 1}$ requires both T_u and $T_{u'}$ to contain at least $\kappa_{\min}^2$ supergraph nodes, equivalently at least $\kappa_{\min}^2$ level-1 districts each, so that every superdistrict bundles a non-trivial number of level-1 districts (Section 4). We define

$$\text{fac}^\ell(T_u) = \begin{cases} 1, & \ell = 2 \text{ and } |V(T_u)| \geq \kappa_{\min}^2 \text{ and } |V(T_{u'})| \geq \kappa_{\min}^2, \\ 1, & \ell = 1 \text{ and } V(T_u) \cap F^1 \neq \emptyset \text{ and } V(T_{u'}) \cap F^1 \neq \emptyset, \\ 0, & \text{otherwise,} \end{cases}$$

for the level-specific admissibility predicate. *The feasibility function* for a subtree T_u at level ℓ is then defined as

$$\phi^\ell(T_u) := \begin{cases} c \cdot \text{fac}^\ell(T_u), & \text{if } d(T_u) \in I^{(r)}(c) \text{ for some } c \in [c_{\min}^\ell, c_{\max}^\ell], \\ 0, & \text{otherwise,} \end{cases} \quad (27)$$

where $I^{(r)}(c)$ is the debt-corrected window (25)- (26). The feasibility function therefore returns the unique potential capacity x when T_u is admissible, and zero otherwise. Uniqueness of c requires that debt-corrected windows $\cdot I^{(r)}(c)$ are pairwise disjoint, which holds by Claim A.4 provided $\epsilon^\ell < 1/(2c_{\max}^\ell)$. This condition is met for typical FalCom parameters such as $c_{\max}^1 = 2$, $c_{\max}^2 = 6$ and $\epsilon^1 = \epsilon^2 = 0.05$.

Theorem A.1 guarantees that each $I^{(r)}(c)$ is non-empty at every recursion step. However, the demands of subtrees form a discrete set, so a non-empty real interval need not contain any achievable demand. Theorem A.7 gives a sufficient condition $\tau^{(r)} \geq \max_{v \in H^{(r-1)}} d(v)$, where $d(v)$ is the demand of node v . Since smaller ϵ^ℓ yields districts whose demands are closer to global demand target $\bar{d}$, it is natural to ask how small ϵ^ℓ can be made. Theorem A.8 provides a necessary lower bound on ϵ^ℓ .

722 **Choosing an admissible subtree.** Let $\mathcal{T}^{(r-1)}$ be the set of all subtrees of $T^{(r-1)}$ and let
723 $\mathcal{E}_{r-1}^\ell = \{T_u \subset T^{(r-1)} : \phi^\ell(T_u) > 0\} \subset \mathcal{T}^{(r-1)}$ denote the set of admissible subtrees. The algorithm
724 selects a subtree $T_u \in \mathcal{E}_{r-1}^\ell$, sets the extracted district $D_r := V(T_u)$, and assigns $c^\ell(D_r) := c$, where
725 c is the corresponding unique potential capacity c .

726 A simple design would sample T_u uniformly from $\mathcal{E}_{r-1}^\ell$, generating a contiguous, capacity-demand
727 balanced district first and only then assigning a facility within it. However, district shapes are
728 geometrically blind, and this would create a one-way dependency where the district shapes completely
729 determine which candidate sites remain available. An early cut may isolate a candidate that would
730 otherwise have been excellent choice. We instead bias the cut selection toward subtrees whose
731 induced districts admit good candidates from the outset. Concretely, we sample $T_u \in \mathcal{E}_{r-1}^\ell$ with
732 probability proportional to a candidate-awareness score $\psi^\ell(T_u)$ defined in Section 5.5, which favors
733 district shapes that preserve well-located candidates.

734 Algorithm 4 wraps Algorithm 5 with repeated spanning tree sampling until an admissible subtree
735 is found, or rejects after M attempts.

Algorithm 4: Capacitated Tree Cut

Input: Residual graph $H^{(r-1)}$, remaining capacity $c_{H^{(r-1)}}$, max capacity $c_{\max}^\ell$, set of debt
corrected windows $I^{(r)}$, max attempts M

Output: Extracted district D_r with capacity assignment $c^\ell(D_r)$ or Rejection

```

1  for  $m = 1$  to  $M$  do
2     $T^{(r-1)} \leftarrow$  random spanning tree of  $H^{(r-1)}$  ;           // via Wilson's algorithm
3    Choose a random root  $v^*$ ;
4     $\mathcal{E}_{r-1}^\ell \leftarrow \text{FindFeasibleCuts}(T^{(r-1)}, c_{H^{(r-1)}}, c_{\max}^\ell, I^{(r)}, v^*)$  ;           // Algorithm 5
5    if  $\mathcal{E}_{r-1}^\ell \neq \emptyset$  then
6      Select  $T_u$  from  $\mathcal{E}_{r-1}^\ell \propto$  candidate-awareness score  $\psi(T_u)$  ;           // Section 5.5
7       $D_r \leftarrow V(T_u)$ ;
8       $c^\ell(D_r) \leftarrow c$ , where  $c$  is the unique potential capacity of  $D_r$ 
9      return  $D_r, c^\ell(D_r)$ 
10   end
11 end
12 return Rejection ;           // no admissible cut after  $M$  attempts

```

737 The retry loop in Algorithm 4 serves as an internal rejection step. A given spanning tree may
738 fail to admit any admissible subtree for two distinct reasons. First, the sampled spanning tree
739 happens to place no admissible subtree, even though the other trees of the same residual would.
740 Second, the residual graph itself admits no admissible subtree under the current debt state, which is
741 structurally impossible whenever $\tau^\ell \geq \max_{v \in H^{(r-1)}} d(v)$ by Theorem A.7. Enforcing this granularity
742 floor $\tau^\ell \geq \max_{v \in H^{(r-1)}} d(v)$ at the start of a run rules out structural failures, thus rejection rates
743 reflect only tree-sampling variance. The first failure mode is addressed by resampling a spanning
744 tree, if the loop exits without finding an admissible subtree, the entire FalCom proposal is rejected
745 and the chain remains at state $s^{(i)}$. We use $M = 1000$ in our experiments, since most successful
746 spanning trees admit an admissible subtree and M only controls worst-case wall-clock time.

747 **Cut mode.** Algorithm 5 evaluates feasibility for every subtree $T_u \subset T^{(r-1)}$ at every capacity
748 $c \in [c_{\min}^\ell, c_{\max}^\ell]$. Both T_u and its complement $T_{u'}$ are candidates for extraction, so each admissible
749 component contributes an entry to $\mathcal{E}_{r-1}^\ell$. However, we do not require both sides of the cut being

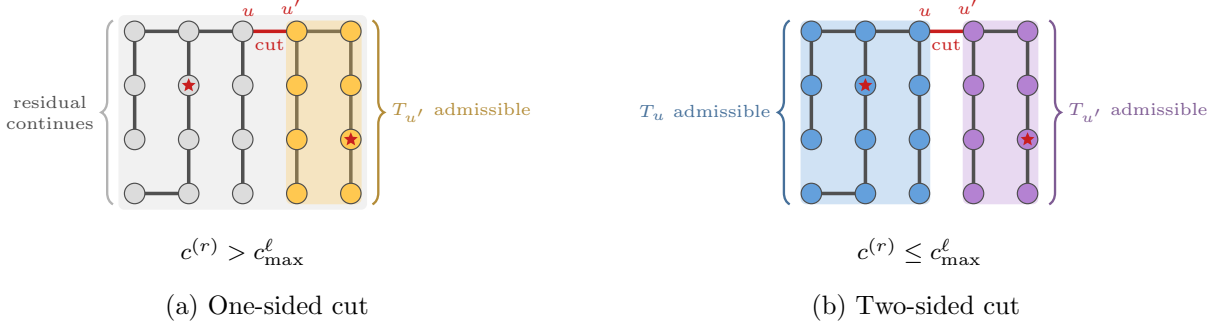


Figure 4: Capacitated tree cut modes. **(a)** One-sided mode ($c^{(r)} > c_{\max}^{\ell}$): Feasibility must hold for only side $T_{u'}$, i.e., $\phi(T_{u'}) > 0$. The residual continues to the next recursion step. **(b)** Two-sided mode ($c^{(r)} \leq c_{\max}^{\ell}$): Both sides must satisfy the feasibility condition, i.e., $\phi(T_u) > 0$ and $\phi(T_{u'}) > 0$. Stars (★) mark candidate facilities.

feasible, because having $\phi(T_u) > 0$ would be enough as long as $c^{(r)} > c_{\max}^{\ell}$. This guarantees a district extraction in the current recursion step, $H^{(r)} = H^{(r-1)}[V(T_{u'})]$ is considered the residual graph in the next recursion step ($r + 1$), and the debt mechanism guarantees that subsequent extractions can still be balanced by A.1. This one-side feasibility mode is called *one-sided cut*, see Figure 4a.

On the other hand, the spanning tree $T^{(r-1)}$ itself is admissible whenever $c_{H^{(r-1)}} \leq c_{\max}^{\ell}$ and $\phi(T^{(r-1)}) > 0$, in which case extracting all remaining nodes as a single district terminates the recursion. For this reason, we consider a *two-sided cut* mode, requiring both $\phi(T_u) > 0$ and $\phi(T_{u'}) > 0$ as shown in Figure 4b, to ensure $c_u + c_{u'} = c_{H^{(r-1)}}$ for the potential capacities c_u and $c_{u'}$ providing local balance for the T_u and $T_{u'}$, respectively. The two-sided cut does not necessarily terminate the recursion, extraction may continue one district at a time as long as $T^{(r-1)}$ is not the subtree selected from $\mathcal{E}_{(r-1)}^{\ell}$, but it secures $T_{u'}$ being fully extracted in the next recursion step ($r + 1$). The debt invariant $|\delta^{(r-1)}| \leq \tau^{\ell}$ proved in Appendix A guarantees that an admissible two-sided cut exists if $\tau^{\ell} \geq \max_{v \in H^{(r-1)}} d(v)$ holds.

Algorithm 5 enumerates all admissible subtrees for a given spanning tree via a single dept-first traversal that incrementally accumulates subtree demands, total work is $O(|H^{(r-1)}|)$ per spanning tree.

Algorithm 5: Find Feasible Cuts

Input: Spanning tree $T^{(r-1)}$ with root v^* , remaining capacity $c_{H^{(r-1)}}$, max capacity $c_{\max}^\ell$, debt-corrected windows $I^{(r)}$

Output: Set of admissible subtrees $\mathcal{E}_{r-1}^\ell$

```

1  $\mathcal{E}_{r-1}^\ell \leftarrow \emptyset$ ;
2 for each subtree  $T_u \subset T^{(r-1)}$  do
3    $T_{u'} \leftarrow T^{(r-1)} \setminus T_u$ ;
4   if  $c_{H^{(r-1)}} > c_{\max}^\ell$  ; // one-sided
5   then
6     if  $\phi(T_u) > 0$  then  $\mathcal{E}_{r-1}^\ell \leftarrow \mathcal{E}_{r-1}^\ell \cup \{T_u\}$ ;
7     else if  $\phi(T_{u'}) > 0$  then  $\mathcal{E}_{r-1}^\ell \leftarrow \mathcal{E}_{r-1}^\ell \cup \{T_{u'}\}$ ;
8   else
9     if  $\phi(T_u) > 0$  and  $\phi(T_{u'}) > 0$  then  $\mathcal{E}_{r-1}^\ell \leftarrow \mathcal{E}_{r-1}^\ell \cup \{T_u\}$  ; // two-sided
10    ;
11  end
12 end
13 return  $\mathcal{E}_{r-1}^\ell$ 

```

A few additional remarks on the design parameters and comments:

- *Granularity floor.* As shown in Appendix A, the choice of $c_{\max}^\ell$ is not free, feasibility requires $\tau^\ell \geq \max_{v \in G^1} d(v)$, where $d(v)$ is the demand of node v . Otherwise, the debt-corrected window would be narrower than the demand of a single node, in which case the smallest possible district $D = \{v\}$ already violates the local balance constraint $d(D) \notin I^{(r)}$.
- *Minimum number of level-1 districts per superdistrict.* The capacity bound $c_{\min}^2 \geq 2c_{\min}^1$ constrains the number of teams in a superdistrict, not the number of level-1 districts it contains; a single level-1 district with $c^1 = c_{\min}^2$ already satisfies it. To force every superdistrict to bundle at least two level-1 districts, matching the MILP constraint (8), the supergraph cut must also satisfy a district-count bound $\kappa_{\min}^2 \in \mathbb{Z}_{\geq 1}$: every admissible level-2 subtree T_u and its complement $T_{u'}$ must each contain at least $\kappa_{\min}^2$ supergraph nodes. With $\kappa_{\min}^2 = 1$ the bound is vacuous; with $\kappa_{\min}^2 = 2$ the level-2 hierarchy becomes non-trivial, preventing the degenerate state in which a superdistrict collapses to a single level-1 district. The level-2 admissibility predicate of Section 5.3 encodes this size constraint directly.
- *Coupling across levels.* The hierarchical consistency bound in Theorem (A.10) links the feasibility thresholds at the two levels and guides the joint choice of ε^1 and ε^2 .
- *Wilson's algorithm.* Wilson's loop-erased random walk with uniform transition probabilities samples spanning trees uniformly at random. On planar graphs of bounded average degree, the expected runtime is near linear in $|H^{(r-1)}|$. Candidate awareness enters through the cut selection probability, not through the spanning tree distribution itself. Biasing the tree distribution to further smooth district boundaries with respect to a candidate-awareness score is a natural extension we leave to future work.

The next section describes the geometric quantities used by the candidate-awareness score ψ and specifies how facilities are assigned to the resulting districts.